Mullvad VPN Documentation

What it is, what it does and why it was chosen for this repository

Technical documentation · British English
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1. Introduction

Mullvad VPN is a virtual private network (VPN) service known for its focus on privacy. This documentation explains what it is, what it does, why it was chosen for this repository and how it contributes to exploring the Surface Web and, as a complement, the dark web.

The text describes the tool's role within the project and does not replace the manufacturer's official material. Prices, features and interfaces may change; confirm current details on Mullvad's official website.

Tools in the repository

Tool	Role in this repository
Mullvad VPN	Network layer: privacy and security of the connection when exploring the Surface Web and, as a complement, the dark web
ChatGPT	Organisation of the repository: folder structure, naming and categorisation
Gemini	Choice of the websites that make up the repository
Claude	Writing documentation and reports

2. What Mullvad VPN Is

A VPN creates an encrypted tunnel between your device and a server operated by the VPN provider. From then on, the websites you visit see the IP address of the server rather than that of your connection, and your internet provider sees only that encrypted traffic is going to the VPN.

Mullvad is run by a Swedish company and has been in operation since 2009. It stands out from other commercial VPNs by using an anonymous account system rather than registration with an email address and password.

- **Anonymous account**: signing up generates a random 16-digit account number, with no email address, name or password. That number is the only identifier; anyone who loses it cannot recover the account.
- **No activity logging**: the service states that it keeps no activity logs and has undergone independent audits.
- **Simple, fixed pricing**: a single plan at a fixed price per 30-day period (around 5 euros at the time of writing), with no promotional tricks.
- **Flexible payment**: accepts methods such as card, cryptocurrency and cash sent by post, which allows payment without linking a name to the account.
- **Open source**: the apps are open source, which makes third-party verification easier.

3. Function and Main Features

- **Traffic encryption**: protects data between the device and the VPN server, especially useful on public Wi-Fi.
- **Hiding your real IP**: sites and services see Mullvad's server IP.

- **WireGuard**: a modern, fast protocol with a lean code base, the service's default, with a tunnel designed to resist quantum-computing attacks.
- **Multihop**: routes the connection through two servers in different places, making correlation harder.
- **Kill switch and lockdown mode**: stop traffic if the VPN drops, preventing your real IP from leaking.
- **Obfuscation**: modes that disguise VPN traffic so that it works on networks that block VPNs.
- **RAM-only servers**: data is not written to disk and disappears when a server restarts.
- **DNS-based ad and tracker blocking**, split tunnelling and several simultaneous devices (up to 5).
- **Mullvad Browser**: a free browser, developed in partnership with the Tor Project, which reduces fingerprint-based tracking.

Note: a VPN shifts trust from your internet provider to the VPN provider. This is why the service's reputation, transparency and audits weigh so heavily in the choice.

4. Why Mullvad Was Chosen

The choice is a project decision, and respectable alternatives exist. The reasons for selecting Mullvad for this repository were:

- **Privacy from sign-up**: it requires no personal data, which suits a project that researches sites of very different kinds without tying the activity to an identity.
- **Transparency**: independent audits, open-source apps and clear communication about how it works.
- **Simplicity**: a single price, no confusing plans, and easy-to-use apps on the main platforms.
- **Features suited to research**: multihop, lockdown mode, obfuscation and detailed server selection by country.
- **Fit with the privacy ecosystem**: compatibility with Mullvad Browser and good coexistence with Tor Browser, documented in this series.

5. Contribution to Exploring the Surface Web

The repository gathers 68 Surface Web sites in twelve categories, from social networks to intelligence bodies. When visiting them, Mullvad contributes in several ways:

- **Protection on insecure networks**: encrypts the connection on public Wi-Fi and unfamiliar networks.
- **Less exposure to the internet provider**: the provider no longer sees which sites are being accessed.
- **IP not exposed to sites**: makes it harder to link a visit to the user's real address.
- **Regional comparison**: by choosing servers in different countries, it is possible to see how news portals, government services and platforms behave in each region, noting differences in language, content and availability.

- **Research with less location-based profiling**: reduces the effect of a home IP address on adverts and results.

What a VPN does	What a VPN does not do
Encrypts the route to the VPN server	Does not make you anonymous on its own
Hides your real IP from sites	Does not stop tracking by cookies, logged-in accounts or browser fingerprinting
Hides destinations from your internet provider	Does not protect against phishing, malware or scams
Lets you choose your apparent location	Does not remove the need to trust the VPN provider

6. Contribution to Exploring the Dark Web

The dark web is reached through **Tor Browser**, and it is Tor, not the VPN, that provides layered anonymity and access to .onion addresses. Mullvad acts as a complementary layer:

- **Hiding Tor use from the local network and provider**: by connecting the VPN before opening Tor, the provider sees only traffic to Mullvad (a setup known as Tor over VPN).
- **Additional protection on untrusted networks**, such as public Wi-Fi.
- **A coherent toolchain**: Mullvad Browser and Tor Browser share the same lineage of fingerprinting protection.

Warning: the VPN does **not** provide access to .onion services, and combining a VPN with Tor does not make Tor more anonymous; it only changes who sees what. The Tor Project states that a VPN is not necessary to use Tor and that combining them requires care. To hide Tor use on networks that block it, Tor Browser's own bridges, such as Snowflake, are an alternative. See the Tor Browser documentation in this series.

7. Getting Started

1. Go to Mullvad's official website (mullvad.net) and generate an account number. Keep it somewhere safe, such as a password manager.
2. Add time to the account using the payment method of your choice.
3. Download and install the official app for your system.
4. Open the app, sign in with the account number and choose a server.
5. Turn on the kill switch or lockdown mode in the settings.
6. Test the connection and check, on the service's official website, that the IP and DNS appear as Mullvad's and that there is no leak.

8. Limitations and Risks

- **A VPN is not total anonymity**: logged-in accounts, cookies and browser fingerprinting still identify you.
- **Trust in the provider**: the guarantees depend on Mullvad's policies and audits.

- **Losing the account number**: with no email address, there is no recovery.
- **Performance**: speeds may drop and latency may rise.
- **Blocking**: some services restrict access from VPN IP addresses.
- **Legislation**: using a VPN is lawful in most countries, but some impose restrictions; respect local law and each service's terms.

9. Good Practice

- Keep the app and operating system up to date.
- Use the kill switch and periodically confirm there are no IP or DNS leaks.
- Combine the VPN with a privacy-focused browser, such as Mullvad Browser, and with good password habits and two-step verification.
- Do not log in to personal accounts if the aim is to browse without being identified.
- Use Tor Browser, not just the VPN, to access .onion services.

10. Conclusion

Mullvad VPN was chosen for combining privacy from sign-up, transparency and simplicity. In the repository, it strengthens the network layer for exploring the Surface Web more safely and serves as an optional complement to Tor Browser on the dark web. It is an important piece, but it does not settle the question of anonymity on its own: the outcome depends on the set of tools and on the habits of those who use them.